ROOT CLASSES - MATHEMATICS

Class 10th NCERT Study Material & Formulas

Chapter 6: Triangles (Tribhuj)

Key Concepts & Important Formulas:

* Similarity of Triangles: Two triangles are similar if corresponding angles are equal and corresponding sides are proportional.

* Basic Proportionality Theorem (Thales Theorem): If a line is drawn parallel to one side of a triangle intersecting other two sides, it divides them in the same ratio.

* Converse of BPT: If a line divides any two sides of a triangle in the same ratio, then the line is parallel to the third side.

* Criteria for Similarity: AAA / AA criterion, SSS criterion, SAS criterion.

Solved Exam Examples & Practice:

Example: In triangle ABC, DE || BC such that AD = 1.5 cm, DB = 3 cm, AE = 1 cm. Find EC.
By Thales Theorem: AD/DB = AE/EC => 1.5/3 = 1/EC => EC = 2 cm.